Math 340: Elementary Matrix and Linear Algebra
MIDTERM ONE

Tuesday, October 11, 2022

Please circle your instructor and your Discussion Session (start time given) in the table below.

Dr. Skylar Grey	Dr. Boya Wen	Dr. Ziquan Yang
Zaidan Wu, T 3:30	Alex Pahlow, M 12:05	Connor Simpson, T 7:45
Zaidan Wu, T 4:35	Ugur Cin, M 12:05	Asvin Gothandaraman, T 8:50
Lauren Neurdorf, R 3:30	Ugur Cin, W 12:05	Connor Simpson, R 7:45
Hyun Jong Kim, R 9:55	Pubo Huang, W 1:20	Asvin Gothandaraman, R 9:55
Hyun Jong Kim, T 9:55	Woojin Kim, M 3:30	
Hyun Jong Kim, R 8:50	Woojin Kim, W 3:30	
	Ugur Cin, W 4:35	
	Woojin Kim, M 2:25	

Dr. Kaitlyn Phillipson	Dr. Zane Li
Emma Thomas, R 11:00	Jacob Denson, M 1:20
Emma Thomas, R 12:05	Jacob Denson, M 2:25
Emma Thomas, R 1:20	Jacob Denson, W 2:25

Name: ______________________________ Wisc email: ___________________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of nine problems for a total of 50 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (5 points, 1 point each) Are each of the following true or false statements? Circle your answer. Justification is not necessary.

a.) **T** **F** If A is a 3×4 matrix, and $\mathbf{x}$ is a 4×1 column vector, then $A\mathbf{x} = \mathbf{0}$ only has the trivial solution $\mathbf{x} = \mathbf{0}$.

b.) **T** **F** If $\det(A) = -3$, then A is an invertible (also called nonsingular) matrix.

c.) **T** **F** For any two square $n \times n$ matrices A and B, $(A+B)(A-B) = A^2 - B^2$.

d.) **T** **F** If A is a 7×8 matrix and B is an 8×8 matrix, then AB is a 7×7 matrix.

e.) **T** **F** For any two invertible $n \times n$ matrices A and B, $(AB)^{-1} = B^{-1}A^{-1}$.

2. (5 points) The linear system $CA^T\mathbf{x} = \mathbf{b}$ is such that A and C are nonsingular (also called invertible), with

$$A^{-1} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}, \; C^{-1} = \begin{bmatrix} 3 & 1 \\ -2 & -1 \end{bmatrix}, \; \mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Find the solution $\mathbf{x}$. Fully justify your answer. (Hint: You do **not** need to find A and C to solve this problem.)

3. (6 points total) Let

$$D = \begin{bmatrix} 3 & -2 \\ 1 & 3 \\ 5 & 1 \end{bmatrix}, E = \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}, F = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 0 & 4 \\ -3 & 1 & 1 \end{bmatrix}, G = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 0 & 2 \end{bmatrix}$$

be four matrices. Compute the following (if possible). If the computation is not possible, explain why not in one sentence.

(a) (2 points) $(DE)^T$

(b) (2 points) $-2DF$

(c) (2 points) $GD - E$

4. (3 points) Let $A = \begin{bmatrix} -1 & 2 & 0 \\ 4 & \alpha & 1 \\ 0 & 3 & 1 \end{bmatrix}$. Find the value(s) of α which would make A *noninvertible* (also called *singular*).

5. (4 points) Let $C = \begin{bmatrix} 0 & 1 & -1 \\ 0 & -2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$. Compute C^{-1}.

6. (7 points total) Consider the linear system $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 4 \\ 1 & -3 & 8 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -10 \\ -18 \\ 3 \end{bmatrix}$

   (a) (3 points) Write out the system of equations in x, y, z.

   (b) (4 points) Through elementary row operations, the augmented matrix $[A : \mathbf{b}]$ is reduced to

$$\left[\begin{array}{ccc:c} 1 & -1 & 3 & -10 \\ 2 & 0 & 4 & -18 \\ 1 & -3 & 8 & 3 \end{array}\right] \to \left[\begin{array}{ccc:c} 1 & -1 & 3 & -10 \\ 0 & 2 & -2 & 2 \\ 0 & 0 & 3 & 15 \end{array}\right].$$

   Use this reduction to finish solving the equation $A\mathbf{x} = \mathbf{b}$.

7. (6 points) Solve the following linear system by using Gauss-Jordan Elimination (i.e. finding the reduced row echelon form). If it has infinitely many solutions, provide an explicit parameterized form for these solutions. If it has no solutions, state this clearly. Show all work.

$$\begin{array}{rcrcrcl} x & + & 2y & - & 2z & = & 0 \\ 2x & + & 3y & - & z & = & 0 \\ -3x & - & 3y & - & 3z & = & 0 \end{array}.$$

8. (7 points total)

    a. (3 points) Let A, B, and C be 3×3 matrices with $\det(A) = 4, \det(B) = 6$, and $\det(C) = -3$. Compute $\det(2A^{-1}B^TC)$. Show all work.

b. (4 points) Let D be the matrix obtained from E after the following row operations:

$\mathbf{r}_1 \longleftrightarrow \mathbf{r}_2$
$3\mathbf{r}_1 + \mathbf{r}_3 \to \mathbf{r}_3$
$(-4)\mathbf{r}_2 + \mathbf{r}_3 \to \mathbf{r}_3$
$\frac{1}{2}\mathbf{r}_4 \to \mathbf{r}_4$.

If $\det(D) = 10$, find $\det(E)$. Show all work.

9. (7 points) Consider the matrix transformation (also called a linear transformation) $f(\mathbf{u}) = A\mathbf{u}$, where

$$A = \begin{bmatrix} -1 & 4 & 3 \\ 0 & 3 & 0 \\ 0 & 6 & -3 \end{bmatrix}$$

a. (2 point) Compute $f(\mathbf{u})$, where $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$.

b. (5 points) Is $\mathbf{w} = \begin{bmatrix} 19 \\ 6 \\ 0 \end{bmatrix}$ in the range/image of f? Fully justify your answer.

Overflow work space for Problem 9 (if needed):

Scratch paper page!!

Back of scratch paper page!